

DD_EM-01 - Franchise Management

Version: v1.0

Status: Draft for review

Bundle: 10_other

Depends on:

- DD_CROSS-00-System_Traceability_and_Implementation_Map-v1.0
- DD_API-00-API_Catalog_and_Gateway_Standards-v1.0
- FOUNDATION_AUTH
- FOUNDATION_KAFKA
- FOUNDATION_FILE_STORAGE
- DD_RLM-CFG-01-HomePass_Configuration-v1.0
- DD_ILM-CFG-02-Tech_Region_Registry-v1.0
- DD_EM-02-Contractor_and_Staff_Registry-v1.0
- DD_SALES-01-Lead_Territory_and_Franchise_Sales_Management-v1.0
- DD_SIP-01-Package_Management-v1.0
- DD_REP-01-Reporting_Data_Mart_and_Dashboard_API-v1.0

Future dependencies:

- DD_EM-CFG-03-RBAC_Catalog
- DD_EM-CFG-04-Approval_Workflow_Catalog
- DD_SIP-03-Discount_Assignment
- DD_SIP-05-Promotional_Campaigns

1. Purpose

This DD defines the Franchise Management process for SOPHIX BSS.

EM-01 is the business source of truth for commercial franchises: who the franchise is, where it operates, what commercial scope it has, whether it is active, and how other modules resolve a customer, HomePass, sale, ticket, or report to a franchise.

The goal is not just to store a list of franchises. The goal is to make franchise context consistent across:

- HomePass and serviceability;
- outdoor sales assignment;
- backoffice filters and user scope;
- order and subscription attribution;
- billing and reporting segmentation;
- contractor/franchise staff identity;
- future campaigns, discounts, and commissions.

Kenya/WIK is the first deployment. The design must also support future operators where franchises may be called zones, branches, regions, dealers, partners, territories, or commercial areas.

2. What This Module Is For

Use EM-01 when the system needs to answer questions like:

Question	EM-01 answer
Which franchise owns or commercially covers this HomePass?	Resolve by HomePass/franchise assignment.
Which franchise should see this lead in the sales app?	Resolve from territory/HomePass/franchise relationship.
Which franchise should appear on an order/subscription for attribution?	Resolve at order capture and persist downstream.
Which franchise manager can supervise this sales team?	Resolve franchise users/teams through EM-02 and RBAC.
Is this franchise active and allowed to sell?	Read franchise lifecycle status and effective dates.
What are the boundaries of this franchise?	Read configured boundary polygons/metadata.
How should reporting group performance?	Use franchise code and hierarchy from EM-01.

EM-01 should be implemented before the next round of frontend and commercial patches, because many documents already refer to franchise context without a single owner.

3. Scope

Included:

Capability	Purpose
Franchise master	Stable franchise id, code, name, operator, lifecycle, type, and manager references.
Franchise lifecycle	Draft, active, suspended, retired, with audit and downstream events.
Franchise boundaries	Optional commercial service area polygons and overlap/exclusion zones.
Franchise HomePass assignment	Which franchises can commercially sell or support a HomePass.
Franchise external references	Legacy/BH/import codes and external partner references.
Franchise contact and escalation metadata	Lightweight business contact points; staff identity remains in EM-02.
Franchise resolution APIs	Resolve franchise by id, code, HomePass, territory, or user scope.
Franchise events	Publish changes for SALES, RLM, FUL, billing, reporting, and frontend caches.

Out of scope:

Out of scope	Owner
Contractor company capacity and slot commitments	EM-02
Individual contractor/staff roster	EM-02
Sales leads and sales activities	SALES-01
Sales territories and route planning	SALES-01, using EM-01 franchise references
HomePass technical lifecycle/status/network path	RLM-CFG-01
Tech region definitions	ILM-CFG-02 / RLM-CFG-01 depending existing deployment split
Package/product definition	SIP-01 / PLM catalogs
Per-franchise campaign/discount assignment	Future SIP-03/SIP-05
Pricing calculation	BIL / SIP / PLM
Legal contract repository	External contract system or future procurement DD
Commission payout	Future commission/finance DD if required

4. Ownership Boundaries

4.1 EM-01 vs RLM-CFG-01

RLM-CFG-01 currently contains franchise references because HomePass configuration needs to know which franchises can sell at an address. Keep that integration, but make ownership explicit:

Concern	Owner	Rule
Franchise master identity	EM-01	Franchise code/name/status/type are mastered here.
Franchise commercial boundary	EM-01	Boundary is owned here; RLM may read/cache for map/serviceability.

Concern	Owner	Rule
HomePass technical status and network path	RLM-CFG-01	EM-01 does not own HomePass lifecycle.
HomePass to franchise assignment	EM-01 owns commercial assignment; RLM exposes it with HomePass detail	RLM must not invent unknown franchise codes.
HomePass map rendering	RLM-CFG-01 / frontend	Map APIs can combine HomePass data with EM-01 franchise metadata.

Implementation note: if the existing RLM database already has a franchise table, keep it during migration as a compatibility projection, but do not let RLM become a second source of truth for franchise status or business ownership.

4.2 EM-01 vs EM-02

Concern	Owner	Rule
Franchise business entity	EM-01	The commercial franchise registry lives here.
Contractor legal entity / workforce / availability	EM-02	Contractor capacity, staff, skills, slots, and main handler live in EM-02.
Franchise implemented by a contractor company	EM-01 references EM-02	contractor_id is optional and links to EM-02 when the franchise is operated by a contractor.
Franchise users and teams	EM-02 + RBAC	EM-01 stores references; it does not duplicate users.

4.3 EM-01 vs SALES-01

Concern	Owner	Rule
Franchise master	EM-01	SALES reads franchise info.
Sales territory	SALES-01	Territory can reference one or more franchises.
Lead assignment	SALES-01	Uses EM-01 franchise context and EM-02 user/team data.
Sales attribution	SALES-01	Stores franchise id/code snapshot on conversion event.

5. Core Concepts

Concept	Meaning
Operator	The deployed company/tenant, for example WIK.
Franchise	Commercial/operational area or partner scope used for sales, reporting, supervision, and sometimes installation/support ownership.
Franchise type	Deployment-configured category such as INTERNAL, EXTERNAL_PARTNER, B2C, B2B, MIXED, DIRECT_BRANCH.
Boundary	Optional GeoJSON polygon or metadata describing the franchise commercial footprint.
HomePass assignment	Link between a HomePass and one or more franchises allowed to sell or support it.
Primary franchise	The default franchise used when only one commercial attribution can be stored.
Secondary franchise	Additional franchise in overlap areas, B2B/B2C dual coverage, or transition periods.
Franchise lifecycle	Draft/active/suspended/retired state of the franchise.

6. Data Ownership

EM-01 owns these tables:

Entity	Purpose
em_franchise	Master franchise record.
em_franchise_boundary	Optional commercial boundary polygons and exclusion/overlap zones.
em_franchise_homepass_assignment	Commercial assignment between a franchise and a HomePass.

Entity	Purpose
em_franchise_external_ref	Legacy/import/external ids.
em_franchise_contact	Lightweight franchise business contact/escalation metadata.
em_franchise_lifecycle_event	Immutable audit of lifecycle and important changes.

EM-01 stores references to other modules but does not own their records:

Reference	Owner
homepass_id	RLM-CFG-01
tech_region_code	ILM-CFG-02 / RLM-CFG-01
contractor_id	EM-02
manager_user_id	FOUNDATION_AUTH / EM-02
sales_territory_id	SALES-01

7. Core Tables

7.1 em_franchise

What it is for: stores the stable master record for one franchise.

When to use it: create, activate, suspend, retire, display, filter, or resolve a franchise across the BSS.

Column	Type	Purpose	Sample
franchise_id	TEXT PK	Stable internal id. Use generated sortable id.	fr_01J_NRB_002
operator_code	TEXT	Operator/tenant scope.	WIK
franchise_code	TEXT	Stable business code unique per operator.	WIK-NRB-002
display_name	TEXT	Human-facing name.	Nairobi 002 - Lavington/Kileleshwa
short_name	TEXT NULL	Compact UI/report name.	NRB 002
country_code	TEXT	ISO country code.	KE
primary_city	TEXT NULL	Main city/market.	Nairobi
franchise_type	TEXT	Deployment category.	EXTERNAL_PARTNER
commercial_segment	TEXT	B2C, B2B, MIXED, or operator-defined.	B2C
status	TEXT	DRAFT, ACTIVE, SUSPENDED, RETIRED.	ACTIVE
contractor_id	TEXT NULL	EM-02 contractor if the franchise is operated by a contractor.	contractor_ke_nairobi_a
manager_user_id	TEXT NULL	Auth user id for the primary franchise manager.	kc_user_franchise_mgr_002
default_language	TEXT	Main language for communications/admin defaults.	en-KE
supported_languages_json	JSONB	Languages supported by the franchise.	["en-KE", "sw-KE"]
default_currency	TEXT	Currency for commercial defaults.	KES
effective_from	DATE	Date from which the franchise can be used.	2026-06-01
effective_to	DATE NULL	Optional end date.	null
created_by_user_id	TEXT	Auth user who created the row.	kc_user_admin_1
created_at	TIMESTAMP	Creation timestamp.	2026-05-29T09:10:00Z
updated_at	TIMESTAMP	Last update timestamp.	2026-05-29T09:15:00Z
metadata_json	JSONB	Deployment-specific flags.	{"legacyCluster": "Nairobi-2" }

Constraints:

- Unique (operator_code, franchise_code).
- status must be one of the configured lifecycle statuses.
- effective_to must be null or greater than effective_from.
- Active external franchises should have contractor_id unless the operator config allows manual/unlinked operation.

7.2 em_franchise_boundary

What it is for: stores the commercial boundary or coverage geometry for a franchise.

When to use it: map display, HomePass assignment by polygon, sales territory setup, reporting by commercial area, or overlap review.

Column	Type	Purpose	Sample
boundary_id	TEXT PK	Boundary id.	frb_01J_NRB002_MAIN
franchise_id	TEXT FK	Franchise.	fr_01J_NRB_002
boundary_type	TEXT	PRIMARY_AREA, OVERLAP_AREA, EXCLUSION_AREA, REFERENCE_ONLY.	PRIMARY_AREA
geometry_geojson	JSONB	GeoJSON polygon/multipolygon.	{"type": "Polygon", "coordinates": [...]}
source_system	TEXT	Where the boundary came from. Use neutral provenance values in new data.	LEGACY_IMPORT
source_ref	TEXT NULL	External source reference.	BH-NRB-002-POLY
effective_from	DATE	Start date.	2026-06-01
effective_to	DATE NULL	End date.	null
status	TEXT	DRAFT, ACTIVE, RETIRED.	ACTIVE
created_at	TIMESTAMP	Creation timestamp.	2026-05-29T09:20:00Z
updated_at	TIMESTAMP	Last update timestamp.	2026-05-29T09:20:00Z

Implementation note: geometry can start as JSONB to keep the medium-budget deployment simple. If polygon queries become heavy, migrate to PostGIS with the same API contract.

7.3 em_franchise_homepass_assignment

What it is for: records which franchise can commercially sell/support a HomePass.

When to use it: serviceability result display, sales app assignment, order attribution, or franchise reporting by HomePass.

Column	Type	Purpose	Sample
assignment_id	TEXT PK	Assignment id.	fha_01J_HP_9988
operator_code	TEXT	Operator scope.	WIK
homepass_id	TEXT	RLM HomePass id.	hp_01HX5M2K9P3Q7R4T6W8Y0Z2N4P
franchise_id	TEXT FK	Franchise.	fr_01J_NRB_002
assignment_role	TEXT	PRIMARY, SECONDARY, B2B_ONLY, B2C_ONLY, TEMPORARY.	PRIMARY
assignment_reason	TEXT	Why this assignment exists. Use neutral provenance values in new data.	MIGRATED
effective_from	DATE	Start date.	2026-06-01
effective_to	DATE NULL	End date.	null
active	BOOLEAN	Whether assignment is currently active.	true
created_by_user_id	TEXT	User/system creating assignment.	kc_user_dataops_1
created_at	TIMESTAMP	Creation timestamp.	2026-05-29T10:00:00Z

Constraints:

- A HomePass can have multiple active franchise assignments only when assignment roles differ or operator config allows overlap.
- A HomePass should have at most one active PRIMARY assignment for the same commercial segment.
- homepass_id must resolve through RLM-CFG-01 before assignment is accepted.

7.4 em_franchise_external_ref

What it is for: maps a franchise to legacy or external identifiers.

When to use it: migration from BH, data import/export, reconciliation, integration with partner systems.

Column	Type	Purpose	Sample
external_ref_id	TEXT PK	Row id.	fref_01J_LEGACY_002
franchise_id	TEXT FK	Franchise.	fr_01J_NRB_002
system_code	TEXT	External system.	BH
external_code	TEXT	External identifier.	Nairobi 002
external_name	TEXT NULL	External display name.	Nairobi-2
active	BOOLEAN	Whether mapping is active.	true
created_at	TIMESTAMP	Creation timestamp.	2026-05-29T09:30:00Z

7.5 em_franchise_contact

What it is for: stores lightweight business contacts for a franchise.

When to use it: backoffice display, escalation, data migration, and operator admin. It is not a replacement for EM-02 staff records.

Column	Type	Purpose	Sample
contact_id	TEXT PK	Contact row id.	frc_01J_MGR_002
franchise_id	TEXT FK	Franchise.	fr_01J_NRB_002
contact_role	TEXT	MANAGER, COMMERCIAL_LEAD, TECH_ESCALATION, FINANCE_CONTACT.	MANAGER
display_name	TEXT	Contact name.	Mary Wanjiku
phone_number	TEXT NULL	Phone.	+254700111222
email	TEXT NULL	Email.	mary.wanjiku@example.co.ke
auth_user_id	TEXT NULL	Auth user if known.	kc_user_franchise_mgr_002
active	BOOLEAN	Whether contact is active.	true
created_at	TIMESTAMP	Creation timestamp.	2026-05-29T09:35:00Z

7.6 em_franchise_lifecycle_event

What it is for: immutable audit trail for important franchise changes.

When to use it: show history, audit admin changes, debug downstream event projection, or rebuild reporting facts.

Column	Type	Purpose	Sample
event_id	TEXT PK	Event id.	fle_01J_ACT_002
franchise_id	TEXT FK	Franchise.	fr_01J_NRB_002
event_type	TEXT	Business event type.	FRANCHISE_ACTIVATED
old_status	TEXT NULL	Prior status.	DRAFT
new_status	TEXT NULL	New status.	ACTIVE
actor_user_id	TEXT	User/system actor.	kc_user_admin_1
reason_code	TEXT NULL	Reason.	INITIAL_SETUP
details_json	JSONB	Change details.	{"approvedBy": "kc_user_ops_1"}
created_at	TIMESTAMP	Event timestamp.	2026-05-29T10:05:00Z

8. Franchise Lifecycle

Status	Meaning	Allowed next statuses
DRAFT	Franchise is being configured. It cannot receive new sales or HomePass assignments in production flows.	ACTIVE, RETIRED
ACTIVE	Franchise can be used by sales, HomePass, order attribution, and reporting.	SUSPENDED, RETIRED
SUSPENDED	Franchise remains visible historically but cannot receive new sales or assignments. Existing orders/subscriptions remain linked.	ACTIVE, RETIRED
RETIRED	Franchise is no longer used for new activity. Historical links remain readable.	terminal

Rules:

Rule	Description
EM01-LC-1	Only ACTIVE franchises can receive new HomePass assignments.
EM01-LC-2	Only ACTIVE franchises can be selected during lead conversion/order attribution.
EM01-LC-3	SUSPENDED franchises remain visible in reports and historical customer/order data.
EM01-LC-4	Retiring a franchise requires confirmation that no active HomePass assignment, open lead, pending order, or open WO still depends on it, unless an approved migration plan exists.
EM01-LC-5	Status changes must create a lifecycle event and emit a Kafka event through outbox.

9. APIs

All APIs require Authorization: Bearer <jwt>.

Create/update/transition APIs require Idempotency-Key.

9.1 Franchise Master APIs

API	Purpose	Consumers
POST /api/franchises	Create franchise in DRAFT.	Backoffice admin.
GET /api/franchises?operatorCode=WIK&status=ACTIVE	List/filter franchises.	Backoffice, SALES, RLM, REP.
GET /api/franchises/{franchise_id}	Franchise detail.	Backoffice, module clients.
GET /api/franchises/by-code/{operator_code}/{franchise_code}	Resolve stable business code.	Module clients, imports.
PATCH /api/franchises/{franchise_id}	Update editable master fields.	Backoffice admin.
POST /api/franchises/{franchise_id}/activate	Activate draft franchise.	Backoffice admin, approval flow later.
POST /api/franchises/{franchise_id}/suspend	Suspend franchise from new activity.	Backoffice admin.
POST /api/franchises/{franchise_id}/retire	Retire franchise.	Backoffice admin.

9.2 Boundary APIs

API	Purpose	Consumers
POST /api/franchises/{franchise_id}/boundaries	Add boundary polygon/metadata.	Backoffice admin, import.
GET /api/franchises/{franchise_id}/boundaries	List boundaries.	Backoffice, map UI, RLM.
PATCH /api/franchise-boundaries/{boundary_id}	Update boundary.	Backoffice admin.

API	Purpose	Consumers
POST /api/franchise-boundaries/{boundary_id}/retire	Retire boundary.	Backoffice admin.

9.3 HomePass Assignment APIs

API	Purpose	Consumers
POST /api/franchises/{franchise_id}/homepass-assignments	Assign HomePass to franchise.	Backoffice admin, migration/import, RLM integration.
GET /api/franchises/{franchise_id}/homepass-assignments	List assigned HomePasses.	Backoffice, reporting.
GET /api/homepasses/{homepass_id}/franchise-assignments	Resolve franchises for HomePass.	SALES, FUL, RLM, self-care/backoffice context.
PATCH /api/franchise-homepass-assignments/{assignment_id}	Update role/effective dates.	Backoffice admin.
POST /api/franchise-homepass-assignments/{assignment_id}/retire	End assignment.	Backoffice admin.
POST /api/franchise-homepass-assignments/bulk-import	Bulk load assignments from BH/GIS/import file.	Data ops.

9.4 Resolution APIs

API	Purpose	Consumers
POST /api/franchises/resolve	Resolve franchise from a context payload.	SALES, FUL, BIL, REP.
GET /api/users/{user_id}/franchise-scope	Return franchises user can see/manage.	FE-APP-01, FE-APP-02.
GET /api/franchises/{franchise_id}/summary	Lightweight read model for UI cards.	Backoffice, sales app.

Resolution request sample:

```
POST /api/franchises/resolve
Authorization: Bearer <jwt>
Idempotency-Key: resolve-franchise-lead-01J
Content-Type: application/json

{
  "operatorCode": "WIK",
  "contextType": "LEAD_CAPTURE",
  "homepassId": "hp_01HX5M2K9P3Q7R4T6W8Y0Z2N4P",
  "geo": {
    "lat": -1.2831,
    "lng": 36.782
  },
  "commercialSegment": "B2C",
  "requestingUserId": "kc_user_sales_04"
}
```

Response sample:

```
{
  "resolutionStatus": "RESOLVED",
  "primaryFranchise": {
    "franchiseId": "fr_01J_NRB_002",
    "franchiseCode": "WIK-NRB-002",
    "displayName": "Nairobi 002 - Lavington/Kileleshwa"
  },
  "secondaryFranchises": [],
  "source": "HOMEPASS_ASSIGNMENT",
  "warnings": []
}
```

10. API Samples

10.1 Create Franchise

```
POST /api/franchises
Authorization: Bearer <jwt>
```


Idempotency-Key: create-franchise-wik-nrb-002
Content-Type: application/json

```
{
  "operatorCode": "WIK",
  "franchiseCode": "WIK-NRB-002",
  "displayName": "Nairobi 002 - Lavington/Kileleshwa",
  "shortName": "NRB 002",
  "countryCode": "KE",
  "primaryCity": "Nairobi",
  "franchiseType": "EXTERNAL_PARTNER",
  "commercialSegment": "B2C",
  "contractorId": "contractor_ke_nairobi_a",
  "managerUserId": "kc_user_franchise_mgr_002",
  "defaultLanguage": "en-KE",
  "supportedLanguages": ["en-KE", "sw-KE"],
  "defaultCurrency": "KES",
  "effectiveFrom": "2026-06-01"
}
```

Response:

```
{
  "franchiseId": "fr_01J_NRB_002",
  "franchiseCode": "WIK-NRB-002",
  "status": "DRAFT"
}
```

10.2 Assign HomePass To Franchise

POST /api/franchises/fr_01J_NRB_002/homepass-assignments
Authorization: Bearer <jwt>
Idempotency-Key: hp-franchise-hp01-nrb002
Content-Type: application/json

```
{
  "homepassId": "hp_01HX5M2K9P3Q7R4T6W8Y0Z2N4P",
  "assignmentRole": "PRIMARY",
  "assignmentReason": "MIGRATED",
  "effectiveFrom": "2026-06-01"
}
```

Response:

```
{
  "assignmentId": "fha_01J_HP_9988",
  "homepassId": "hp_01HX5M2K9P3Q7R4T6W8Y0Z2N4P",
  "franchiseId": "fr_01J_NRB_002",
  "assignmentRole": "PRIMARY",
  "active": true
}
```

11. Business Rules

Rule	Description
EM01-R-1	Franchise code is immutable after activation unless an approved migration correction is run.
EM01-R-2	New sales/order attribution can only use ACTIVE franchises.
EM01-R-3	Historical orders/subscriptions/tickets must keep their original franchise snapshot even if the franchise is renamed later.
EM01-R-4	A HomePass must not be assigned to a retired franchise.
EM01-R-5	A HomePass can have multiple franchise assignments only when overlap is explicitly configured.
EM01-R-6	Only one active PRIMARY assignment is allowed for the same HomePass and commercial segment.
EM01-R-7	Suspending a franchise blocks new leads and new order conversions but does not cancel existing subscriptions.
EM01-R-8	Retiring a franchise requires either zero active dependencies or a migration plan approved by the configured approval workflow.
EM01-R-9	EM-01 must publish changes through outbox so RLM, SALES, REP, and frontend caches can refresh.
EM01-R-10	No module should hardcode franchise names like Nairobi 002; use franchise_id or franchise_code.

12. Events

Event	Topic	Meaning	Consumers
FranchiseCreated	sophix.em.franchise-created	Franchise created in draft.	REP, Backoffice cache.
FranchiseActivated	sophix.em.franchise-activated	Franchise became active.	SALES, RLM, FUL, REP.
FranchiseSuspended	sophix.em.franchise-suspended	Franchise blocked from new activity.	SALES, FUL, Backoffice, REP.
FranchiseRetired	sophix.em.franchise-retired	Franchise no longer usable for new activity.	SALES, RLM, REP.
FranchiseBoundaryUpdated	sophix.em.franchise-boundary-updated	Boundary changed.	RLM map/serviceability, SALES map, REP.
FranchiseHomePassAssigned	sophix.em.franchise-homepass-assigned	HomePass assigned to franchise.	RLM, SALES, REP.
FranchiseHomePassAssignmentRetired	sophix.em.franchise-homepass-assignment-retired	Assignment ended.	RLM, SALES, REP.

Event envelope must follow FOUNDATION_KAFKA and DD_API-00 correlation rules:

```
{
  "eventId": "evt_01J_EM01_ACT",
  "eventType": "FranchiseActivated",
  "operatorCode": "WIK",
  "aggregateType": "Franchise",
  "aggregateId": "fr_01J_NRB_002",
  "occurredAt": "2026-05-29T10:05:00Z",
  "correlationId": "corr_01J_ADMIN",
  "payload": {
    "franchiseCode": "WIK-NRB-002",
    "displayName": "Nairobi 002 - Lavington/Kileleshwa",
    "status": "ACTIVE"
  }
}
```

13. Frontend Usage

13.1 FE-APP-01 Backoffice

Backoffice uses EM-01 for:

Area	API
Franchise admin list	GET /api/franchises?...
Franchise detail	GET /api/franchises/{franchise_id}
Create/edit franchise	POST /api/franchises, PATCH /api/franchises/{franchise_id}
Activate/suspend/retire	lifecycle APIs
Boundary map	boundary APIs plus map component
HomePass assignment view	assignment APIs
User scope filters	GET /api/users/{user_id}/franchise-scope

13.2 FE-APP-02 Outdoor Sales / Franchise App

Sales app uses EM-01 for:

Area	API
Agent franchise badge	GET /api/users/{user_id}/franchise-scope
Lead franchise resolution	POST /api/franchises/resolve or SALES-01 backend resolution
Territory/franchise filters	SALES-01 territory APIs plus EM-01 franchise read model
Supervisor scope	User franchise scope API

13.3 Reporting Portal

Reporting uses EM-01 as a dimension:

Report dimension	Source
franchise_id	EM-01
franchise_code	EM-01
franchise_status	EM-01
commercial_segment	EM-01
country_code, city	EM-01

14. Integration With Other Modules

Module	Integration
RLM-CFG-01	Reads franchise master and assignments; exposes franchise refs with HomePass detail.
SALES-01	Reads franchise list/scope; stores franchise attribution on leads/conversions.
FUL-02	Receives franchise attribution when order is created.
SUB-LM-01	Stores franchise snapshot if needed for subscription reporting; does not call EM-01 on every state change.
BIL	Uses franchise only for reporting/pricing lookup where configured; billing must not infer franchise from free text.
REP-01	Consumes events/read model to build franchise dimensions.
EM-02	Provides contractor/staff/team identity for linked franchise operations.
FOUNDATION_AUTH / future RBAC	Provides user identity and franchise-scoped permissions.

15. Developer Implementation Notes

Implementation should be straightforward if built as a CRUD/read-model module with event publishing.

Recommended sequence:

1. Create the em schema tables for franchise, boundary, HomePass assignment, external refs, contacts, and lifecycle events.
2. Implement repositories and service methods with operator-scoped uniqueness checks.
3. Implement create/read/update/lifecycle APIs.
4. Implement HomePass assignment APIs; validate homepass_id through RLM-CFG-01 API, not direct DB joins.
5. Implement franchise resolution service:
 - first by active HomePass assignment;
 - then by active boundary match if GIS/PostGIS is enabled;
 - then by sales territory if SALES-01 provides a territory context;
 - otherwise return UNRESOLVED with warnings.
6. Add outbox events for every lifecycle and assignment change.
7. Add cache/read model for low-latency franchise scope lookup.
8. Patch SALES-01 and frontend docs to reference EM-01 after review.

Do not implement franchise resolution by querying RLM or SALES database tables directly. Use approved internal API clients.

16. Infrastructure And Deployment

For a budget-conscious medium deployment:

Component	Recommendation
Runtime	Deploy inside the existing EM/business-services pod. EM-01 is low traffic and does not need its own deployment at first.
Database	PostgreSQL em schema in the shared business DB cluster.
Cache	Redis cache for active franchise list and user franchise scope. TTL 5-15 minutes; invalidate on events.
Kafka	Use transactional outbox for franchise events.
Camunda	Not required for v1 unless the operator wants approval workflows immediately. Lifecycle APIs can be synchronous with audit.
Drools	Not required for v1. Boundary/overlap rules are simpler as service validation.
File storage	Optional for boundary import files or supporting documents.
Observability	Alert on outbox backlog, failed HomePass assignment validation, and repeated unresolved franchise resolution.

AWS recommendation:

Concern	AWS guidance
Compute	Run in the shared EM ECS/EKS service; no dedicated node group needed.
Database	Use existing RDS PostgreSQL cluster/schema.
Cache	Use existing ElastiCache/Redis if deployed.
Events	Use MSK or the platform Kafka cluster.
Secrets	Store DB/API credentials in Secrets Manager or SSM Parameter Store.
GIS	If PostGIS is required, enable it in RDS PostgreSQL and add indexes later; do not start with GIS complexity unless boundary auto-resolution is required.

17. Tests

Minimum test coverage:

Test	Expected result
Create franchise with unique code	Row created in DRAFT.
Create duplicate code for same operator	HTTP 409 CONFLICT.
Activate valid draft franchise	Status becomes ACTIVE, lifecycle event and Kafka event emitted.
Assign HomePass to active franchise	Assignment created after RLM validates HomePass exists.
Assign HomePass to retired franchise	HTTP 409 CONFLICT.
Resolve franchise by HomePass	Returns primary franchise.
Resolve unknown HomePass/franchise context	Returns UNRESOLVED, not server error.
Suspend franchise	New sales/order attribution blocked; historical reads still work.
Retire franchise with active dependencies	Blocked unless migration override/approval exists.
Event replay to REP read model	Franchise dimension can be rebuilt.

18. Open Decisions For Review

Decision	Recommendation
Should EM-01 become the canonical franchise master, replacing the RLM franchise catalog?	Yes. RLM should keep HomePass-specific references/projections only.
Should HomePass/franchise assignment be owned by EM-01 or RLM?	EM-01 should own commercial assignment; RLM should expose it with HomePass responses.
Should boundary lookup require PostGIS in phase 1?	No. Start with explicit HomePass assignments and JSONB boundaries. Add PostGIS only if operationally needed.
Should franchise manager/contact be a free text contact or EM-02/Auth user reference?	Support both: user reference when known, free text for migration/legacy.

Decision	Recommendation
Should approval be mandatory for activation/suspension/retirement?	Defer to EM-CFG-04; for v1, implement synchronous lifecycle with audit and add approval hook later.